

WEEKLY INTERNSHIP REPORT

Summer Research Programme 2025

Week Number Week 5 of 8	Report Period 22 / 06 / 2026 → 26 / 06 / 2025
Group & Project Group 2 Project Title: Resource scheduling in 5G	Mentor Name Dr / Prof. Mr. Anil Carie
Intern Name(s) 1. Valluru Komal Sivaram 2. Earla Bhavyanshi 3. Makineni Ruthvik Krishna	This week's title <i>Scenario Variation + Ablation Study</i>

HOW THIS REPORT BUILDS YOUR CONFERENCE PAPER

Section 2 → Related Work (papers you read + Coursera/YouTube concepts learned)

Section 4b → Results section data (every number here becomes a table row in the paper)

Section 5 → Draft sentences copied directly into the paper manuscript

Section 6 → Figures described here become the paper's figures with captions

Section 8 → Limitations section and Future Work paragraph

Fill every section honestly. A blank section = missing paper content.

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WEEK GOAL

Evaluate the performance of the Deep Reinforcement Learning based 5G scheduler under different network load scenarios (low, medium, and high traffic conditions) and compare its behaviour across key KPIs.

This week's goal:

Perform ablation and convergence analysis to study the impact of individual DQN components and validate the stability and effectiveness of the scheduling model.

Key deliverable:
Core Results Table and evaluation outputs including scenario-based KPI comparison, ablation study results, and convergence analysis figures committed to GitHub.

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LEARNING — COURSERA + YOUTUBE

2A. Coursera Progress

Course Name	Modules Completed This Week	Score / Status
Reinforcement Learning Specialization University of Alberta	Modern Practical Deep RL concepts including policy gradient methods, multi-agent reinforcement learning foundations, and practical reinforcement learning applications	Audited / Completed

2B. YouTube Watched This Week

List every video watched. In the 'Key Takeaway' column write what you actually learned — not just the topic.

Video Title	Channel	Duration	Key Takeaway (1 sentence)
Hugging Face Deep RL Course — Unit 3 (Deep Q-Learning)	Hugging Face	1 hr	Learned how Deep Q-Networks use neural networks and experience replay to improve decision-making.
David Silver RL Lecture 6 — Value Function Approximation	DeepMind UCL	1 hr 22 min	Understood value function approximation concepts and convergence behavior in reinforcement learning.

2C. New Concept Learned (explain in your own words)

This week, we learned how reinforcement learning models use function approximation to handle large and complex problems. Instead of storing values for every possible state, the model learns patterns using neural networks. We also understood that training stability and convergence are important because they determine whether the model learns properly or becomes unstable. These concepts helped us understand how algorithms such as DQN can make reliable decisions in our scheduling problem.

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DAILY TASK LOG

Day	Task Completed	File / Output Committed to GitHub	Blocker (if any)
Mon	Evaluated the DQN scheduler under low, medium, and high network load scenarios to study the impact of varying traffic conditions on throughput, latency, and fairness metrics.	results/week5_scenarios.csv (Sivaram)	Difficulty maintaining consistent performance across different traffic loads
Tue	Analysed the QoS-aware reward function and studied how different reward components influence scheduling decisions and overall network performance.	docs/week5_day2.md (Ruthvik)	Balancing multiple QoS objectives within the reward function
Wed	Performed an ablation study by removing selected DQN components and comparing their impact on scheduler performance and learning behaviour.	results/week5_ablation_study.csv (Sivaram)	Identifying the contribution of individual components accurately
Thu	Conducted convergence analysis by observing reward trends and training stability over multiple episodes to evaluate learning behaviour.	results/reward_curve_smoothed.png (Bhavyanshi)	Fluctuations in reward values during training

Day	Task Completed	File / Output Committed to GitHub	Blocker (if any)
Fri	Validated final results and prepared performance comparisons across all experiments for project demonstration and result verification.	results/final_results_table.csv (Bhavyanshi)	Ensuring consistency of evaluation metrics across multiple runs

Weekly Commit Summary

GitHub Repo URL	Total Commits This Week	Most Important Commit (message)
https://github.com/xosivaram2007/5g-scheduling-intern	10+	Added scenario evaluation, ablation study, and convergence analysis for DQN scheduler

4 TECHNICAL WORK THIS WEEK

4A. What I Built / Implemented

This week, our group focused on evaluating the performance and stability of the Deep Reinforcement Learning based scheduler under different network conditions. Sivaram conducted experiments using multiple network load scenarios to analyse the impact of traffic variations on throughput, latency, and fairness metrics.

Ruthvik studied the QoS-aware reward function and analysed how different reward components influence scheduling decisions and overall system performance. The relationship between reward design and Quality of Service requirements was also examined.

Sivaram performed an ablation study by evaluating the contribution of individual DQN components and comparing their effects on scheduling performance. Bhavyanshi carried out convergence analysis by observing reward trends and training stability during the learning process.

By the end of the week, the team completed scenario-based evaluation, ablation studies, convergence analysis, and final result validation, providing a comprehensive assessment of the DQN scheduler's performance and reliability.

4B. Results & Numbers ★ MOST IMPORTANT ★

Method / Model / Config	Metric 1 (Throughput)	Metric 2 (Latency)	Metric 3 (Jain's Fairness Index)	Notes
Round Robin (RR)	Lowest throughput observed under high load	Stable latency	High fairness	Equal cyclic resource allocation
Proportional Fair (PF)	Improved throughput compared to RR	Moderate latency	Maintained fairness	Adaptive scheduling based on throughput
DQN Scheduler	Improved adaptive allocation	Reduced latency observed	Stable fairness performance	Learned scheduling policy through training
DQN without Replay Buffer	C Reduced throughput	Increased latency	Slight Reduction in fairness	Experience replay contributed to stable learning
DQN without Target Network	Less stable performance	Higher latency variation	Lower fairness consistency	Target network improved convergence and stability

Best result this week:

Model = DQN Scheduler

Metric = Overall scheduling performance

Value = Improved scheduling adaptability compared to baseline methods

Compared to baseline: The DQN scheduler demonstrated more adaptive scheduling behaviour than traditional scheduling approaches by learning allocation strategies directly from environment interaction. Experimental evaluation showed improved performance trends and stable learning behaviour across different scenarios.

4C. Key Figure / Plot This Week

[Figure file results/comparison.csv]

Caption draft: *This week focused on model training, evaluation workflow implementation, and benchmark preparation.*

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PAPER CONTRIBUTION LOG

THIS SECTION IS THE HEART OF THE REPORT

Write these sentences THE SAME DAY as the experiment — not on Friday. Memory fades. These sentences are copied into the paper in Weeks 7–8. The better you write here, the easier the paper sprint.
Every week must have at least 3 candidate sentences. Aim for 5.

5A. Candidate Sentences for the Paper

Paper Section	Draft Sentence (write as it would appear in the paper)
Introduction	Resource scheduling performance can vary significantly under different network traffic conditions in 5G systems.
Methodology	The DQN scheduler was evaluated under multiple network load scenarios to analyse scheduling behaviour and adaptability.
Methodology	Ablation experiments were conducted to study the contribution of different DQN components to scheduling performance.
Results	Experimental evaluation showed that the DQN scheduler maintained stable performance across different network scenarios.
Discussion	The analysis demonstrated the importance of training stability and model components for effective reinforcement learning based scheduling.

5B. Gap / Limitation Found This Week

Although the DQN scheduler showed stable performance, the results were influenced by network load conditions and model configuration. Ablation experiments indicated that certain DQN components significantly affect learning stability and scheduling performance, requiring further evaluation and tuning.

5C. Reference Found This Week

#	IEEE Citation (Author, Title, Journal/Conf, Year)	Relevance to Our Work
1	Deep Reinforcement Learning for Wireless Resource Scheduling, IEEE, 2024	Helped analyse scheduling performance, convergence behaviour, and evaluation under different network scenarios

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NEXT WEEK PLAN

Day	Planned Task (specific)	Expected Output / Commit
Mon	Re-run all experiments using multiple random seeds and calculate mean and standard deviation for throughput, latency, and fairness metrics.	Results table with mean $\pm$ standard deviation
Tue	Perform statistical significance testing between DQN and baseline schedulers using throughput and latency results. Add p-values to the comparison table.	Statistical analysis results
Wed	Review additional research papers and prepare the Related Work section covering traditional, machine learning, and DRL-based scheduling approaches.	related_work.md committed
Thu	Generate publication-quality figures including convergence curves, KPI comparisons, fairness analysis, and ablation results at 300 DPI resolution.	Six publication-ready figures
Fri (demo)	Present the final figures and analysis to the mentor, prepare figure captions, and finalize approved results for paper writing.	Approved figures and captions $\checkmark$

Coursera + YouTube Plan for Next Week

Platform	Module / Video Title	When to watch / complete
Coursera	<i>No new modules – focus on evaluation and result validation</i>	<i>Mon to Thurs</i>
YouTube	<i>Matplotlib Tutorial for Beginners – Full Course</i>	<i>Thursday</i>
YouTube	<i>Statistics Fundamentals: p-values and Statistical Significance</i>	<i>Monday</i>

7 SELF-ASSESSMENT

Q1: What surprised you most technically this week?

We were surprised to see how much the performance of the DQN model depended on its individual components and training conditions. The ablation study showed that removing features such as the replay buffer or target network affected learning stability and scheduling performance. This helped us understand that reinforcement learning models require careful design and evaluation to achieve reliable results.

Q2: If you had to redo this week, what would you do differently?

If we repeated this week, we would plan the evaluation process earlier and organize experiment tracking more carefully. This would reduce repeated testing and make it easier to compare scenario, ablation, and convergence results during analysis and paper preparation.

8 MENTOR REVIEW (Mentor fills this section)

Students submit the report by Friday 5pm. Mentor completes this section by Monday 9am before the next week begins.

8A. Checklist

Item	Status	Comment
Key deliverable submitted and committed to GitHub	☒ Yes / ☐ Partial / ☒ No	
Section 4B results table has real numbers	☒ Yes / ☐ Partial / ☒ No	

Item	Status	Comment
Section 5A has ≥ 3 candidate paper sentences	☒ Yes / ☐ Partial / ☒ No	
GitHub commits are clean with meaningful messages	☒ Yes / ☐ Partial / ☒ No	
Friday demo completed	☒ Yes / ☒ No	
Results are reproducible (someone else can run the code)	☒ Yes / ☐ Not yet / ☒ No	

8B. Technical Feedback

8C. Paper Progress Assessment

Paper Section	Status After This Week	Target Week to Complete
Introduction	Not started / Outline / Draft / Final	Week 7
Related Work	Not started / Outline / Draft / Final	Week 6
Methodology	Not started / Outline / Draft / Final	Week 7
Results	Not started / Data collected / Draft / Final	Week 7
Conclusion	Not started / Draft / Final	Week 7
References	Not started / Partial (___ refs) / Complete	Week 8

8D. Go / No-Go for Next Week

☒ PROCEED All deliverables met. Continue to next week's plan.	☐ FIX FIRST Fix specific items before proceeding. See feedback above.	☐ REPLAN Results not credible. Stop and redesign the experiment.
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Decision: Circle one above. If FIX FIRST — list exact items to fix before Monday:

8E. Mentor Signature

Mentor Name	Date Reviewed	Signature

APPENDIX — CUMULATIVE PAPER SKELETON TRACKER

Update this page every week. By Week 6 this should be the first draft of your paper.

HOW TO USE THIS TRACKER

Each row corresponds to a paper section. Paste your best candidate sentences from Section 5A here each week.

By Week 6 you should have 3–5 sentences per row. These become the paper draft in Week 7.

Colour code: Black = confirmed content Grey = draft Red = needs rewrite

A. Introduction

Added in	Content / Draft Sentences
Week 1	Efficient resource scheduling is an important challenge in 5G wireless networks due to increasing user demand and diverse Quality of Service (QoS) requirements. Traditional scheduling algorithms such as Round Robin mainly focus on fairness and do not adapt dynamically to changing network conditions. This project aims to explore AI and deep reinforcement learning techniques for intelligent and adaptive 5G resource allocation.
Week 2	Reinforcement Learning offers a promising approach for adaptive resource allocation in 5G networks by enabling agents to learn scheduling policies through interaction with the environment. Unlike traditional rule-based schedulers, RL-based methods can optimize long-term network performance under dynamic conditions. Modeling the scheduling problem as a Markov Decision Process provides a foundation for intelligent decision-making.
Week 3	Deep Reinforcement Learning enables adaptive resource allocation by allowing agents to learn scheduling policies through interaction with dynamic network environments. This approach provides flexibility compared to traditional fixed scheduling methods and supports intelligent decision-making in complex scenarios.
Week 4	Deep Reinforcement Learning introduces adaptive decision-making capabilities for resource allocation in dynamic wireless environments. Traditional scheduling approaches follow predefined allocation strategies, whereas learning-based methods continuously improve through interaction with the environment. This work evaluates reinforcement learning methods to explore intelligent scheduling for future 5G systems.
Week 5	Resource scheduling performance can vary significantly under different network traffic conditions in 5G systems. Evaluating reinforcement learning schedulers under varying scenarios helps determine their adaptability and robustness. Understanding the contribution of individual model components is also important for developing reliable scheduling strategies.
Week 6	

Week 7	
Week 8	

B. Related Work

Added in	Content / Draft Sentences
Week 1	Existing AI-based scheduling approaches attempt to improve throughput, latency, and fairness compared to traditional heuristic schedulers. A survey paper on deep reinforcement learning for wireless resource allocation was reviewed to understand current research trends and challenges. Reinforcement learning has shown strong potential for adaptive decision-making in dynamic wireless communication environments.
Week 2	Reinforcement Learning has been widely applied to wireless communication problems due to its ability to learn optimal policies without explicit programming. Markov Decision Processes are commonly used to represent resource allocation problems where decisions must be made sequentially over time. Recent studies have shown that RL-based schedulers can improve network efficiency while maintaining fairness among users.
Week 3	Deep Q-Networks (DQN) combine reinforcement learning with neural networks to solve decision-making problems in high-dimensional environments. Existing studies show that DRL methods can improve adaptability and scheduling performance compared to conventional approaches.
Week 4	Recent studies in wireless resource scheduling have explored Deep Reinforcement Learning to optimize network performance metrics including throughput, latency, and fairness. DQN and PPO are commonly adopted approaches due to their ability to learn scheduling behaviour directly from environmental feedback. Benchmark evaluation remains important for validating performance improvements against traditional methods.
Week 5	Recent studies have emphasized the importance of evaluating DRL schedulers under different network conditions to assess their robustness and adaptability. Ablation studies are commonly used to analyse the contribution of individual model components and understand their impact on learning performance. Convergence analysis also plays an important role in validating the stability and reliability of reinforcement learning based scheduling methods.
Week 6	
Week 7	
Week 8	

C. Methodology

Added in	Content / Draft Sentences
Week 1	A baseline Round Robin scheduler was implemented in Python to simulate fair resource allocation among multiple users. The scheduler simulation included 5 user equipments, 10 resource blocks, and 100 scheduling time slots. This implementation established the foundational baseline that will later be compared against AI and reinforcement learning-based scheduling algorithms.
Week 2	A custom Gymnasium environment was developed to model the 5G scheduling problem using states, actions, rewards, and environment transitions. The environment provides a platform for training and evaluating future reinforcement learning agents. Additionally, a Proportional Fair scheduler was implemented as a baseline method for comparison against learning-based approaches.
Week 3	A Deep Q-Network (DQN) agent was implemented using Stable-Baselines3 and trained on a custom scheduling environment. The environment was designed with compatible observation and action spaces to support reinforcement learning-based scheduling experiments.
Week 4	A benchmarking framework was developed to evaluate scheduling performance across Round Robin, Proportional Fair, DQN, and PPO approaches. DQN and PPO agents were trained and evaluated within a custom scheduling environment using selected network performance metrics. The evaluation process measured scheduler effectiveness and adaptability under identical testing conditions.
Week 5	The DQN scheduler was evaluated under multiple network load scenarios to analyse its performance under varying traffic conditions. Ablation experiments were conducted by removing key DQN components to study their contribution to learning performance and scheduling effectiveness. Convergence analysis was performed using reward curves and checkpoint evaluations to assess training stability and learning behaviour.
Week 6	
Week 7	
Week 8	

D. Results

Added in	Content / Draft Sentences
Week 1	The implemented Round Robin scheduler successfully demonstrated cyclic and fair allocation behaviour across all simulated users. The simulation executed successfully over 100 time slots without allocation failures. Initial project KPIs including throughput, latency, and Jain's fairness index were identified for future performance evaluation.

Week 2	Both Round Robin and Proportional Fair schedulers completed 10,000 resource allocations during simulation. Round Robin achieved a Jain Fairness Index of 1.000, while Proportional Fair achieved 0.999, demonstrating near-perfect fairness in both approaches. These baseline results establish a reference point for evaluating future reinforcement learning schedulers.
Week 3	Initial DQN training demonstrated increasing reward behaviour and successful interaction with the scheduling environment. The developed training pipeline established the foundation for future benchmarking against Round Robin and Proportional Fair scheduling algorithms.
Week 4	Benchmark evaluation enabled comparative analysis between conventional scheduling algorithms and reinforcement learning-based approaches. Experimental observations demonstrated differences in scheduling behaviour across throughput, latency, and fairness metrics. The evaluation framework established the foundation for future scenario testing and ablation studies.
Week 5	Experimental evaluation demonstrated that the DQN scheduler maintained stable performance under different network load scenarios. Ablation studies showed that components such as the replay buffer and target network contributed to improved learning stability and scheduling performance. Convergence analysis indicated that the DQN agent achieved stable learning behaviour during training.
Week 6	
Week 7	
Week 8	

E. Discussion / Conclusion

Added in	Content / Draft Sentences
Week 1	The first week of the project provided a strong foundation in Python programming, 5G scheduling concepts, and simulation-based implementation. Although the Round Robin scheduler ensures fairness, it does not consider factors such as channel quality or latency requirements. Future work will focus on reinforcement learning-based schedulers that can optimize multiple network performance metrics simultaneously.
Week 2	The second week successfully established the simulation environment and baseline schedulers required for reinforcement learning experiments. The Proportional Fair scheduler demonstrated that throughput-aware scheduling can maintain high fairness while adapting allocation decisions. Future work will focus on integrating reinforcement learning agents and evaluating their performance against the established baselines.

Week 3	This week established the initial Deep Reinforcement Learning workflow for the project through DQN implementation and training validation. Future work will focus on reward optimization and performance comparison using throughput, latency, and fairness metrics.
Week 4	This week extended the project from baseline scheduling into intelligent scheduling evaluation using reinforcement learning approaches. Although DRL methods demonstrated adaptive behaviour, further optimization and broader evaluation scenarios are necessary before finalizing conclusions. Future work will focus on scenario variation and ablation analysis.
Week 5	This week provided a deeper understanding of the factors affecting reinforcement learning based scheduling performance. Scenario evaluation and ablation studies demonstrated that both network conditions and model components influence learning stability and scheduling effectiveness. Future work will focus on statistical validation, result visualization, and preparation of publication-quality figures for the final paper.
Week 6	
Week 7	
Week 8	

F. References (IEEE format)

Added in	Content / Draft Sentences
Week 1	<i>Survey Paper on Deep Reinforcement Learning for 5G Resource Scheduling, IEEE Xplore, 2024.</i>
Week 2	<i>[1] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018.</i> <i>[2] D. Silver, Reinforcement Learning Course Lectures, DeepMind and University College London, 2015.</i>
Week 3	<i>V. Mnih et al., "Human-level Control Through Deep Reinforcement Learning," Nature, vol. 518, no. 7540, pp. 529–533, 2015.</i>
Week 4	<i>[1] Deep Reinforcement Learning for Wireless Resource Scheduling, IEEE, 2024.</i> <i>[2] Proximal Policy Optimization Algorithms, Schulman et al., 2017.</i>
Week 5	<i>[1] Deep Reinforcement Learning for Wireless Resource Scheduling, IEEE, 2024. This work provided guidance for analysing scheduling performance, convergence behaviour, and evaluation under varying network conditions.</i>
Week 6	

Week 7	
Week 8	